

## hPTM signatures association with orphan genes

Here, I load the supplementary table from Jeromine's hPTM signatures and examine the association between the signatures and orphan genes.

```
library(tidyverse)
library(ggrastr)
library(readxl)
```

```
readxl::excel_sheets("data/dataset_S3_master_tables.xlsx")
```

```
## [1] "Ectocarpus"          "Desmarestia-dudresnayi" "Desmarestia-herbacea"
## [4] "Undaria"            "Scytosiphon"          "Schizocladia"
```

```
Ec_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Ectocarpus") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Ddud_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-dudresnayi") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Dher_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-herbacea") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Und_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Undaria") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Spro_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Scytosiphon") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Si_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Schizocladia") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
```

Note: Scytosiphon is misspelt in the table

```
preprocess_ps_sig_orphan <- function(Species_ps, TPM_filt = 1){
  Species_ps.filt <- Species_ps |>
    tidy::drop_na() |>
    dplyr::mutate(rank = as.numeric(rank)) |>
    tidy::drop_na(rank)
  max_ps <- max(Species_ps.filt$rank)
  if(any(colnames(Species_ps.filt) %in% c("Signature.Female", "Signature.Male"))){
    Species_ps.filt2 <-
      Species_ps.filt |>
      dplyr::mutate(Signature.Female = factor(
        Signature.Female, levels = paste0('S',1:16))) |>
      dplyr::mutate(Signature.Male = factor(
        Signature.Male, levels = paste0('S',1:16))) |>
      dplyr::mutate(n_all.Female = dplyr::n(), .by = Signature.Female) |>
      dplyr::mutate(n_all.Male = dplyr::n(), .by = Signature.Male) |>
      dplyr::mutate(TPM = max(TPM.Female, TPM.Male), .by = gene_id) # max since we are intere
  }else{
    Species_ps.filt2 <-
      Species_ps.filt |>
      dplyr::mutate(Signature = factor(
        Signature, levels = paste0('S',1:16))) |>
```

```

    dplyr::mutate(n_all = dplyr::n(), .by = Signature)
  }
Species_ps.filt3 <- Species_ps.filt2 |>
  dplyr::mutate(orphan = dplyr::case_when(
    rank == max_ps & TPM > TPM_filt ~ "orphan",
    .default = "nonorphan"
  ))
  # dplyr::filter(rank == max_ps)
return(Species_ps.filt3)
}

color_table <-
  tibble::tibble(
    Signature = paste0('S',1:16),
    Colour = c("#a31d21", "#e43a30", "#f36f57", "#f8a68d", "#fddfd1", "#1b733b",
"#2daf66", "#6ec283", "#aed9b1", "#9392c8", "#bfbfd", "#e9eef8",
"#2a74bb", "#61abd7", "#afd1e5", "#999999")
  )

plot_style <- function(p_in){
  p_out <-
    p_in +
    ggplot2::rasterise(ggplot2::geom_jitter(width = 0.3, alpha = 0.1)) +
    ggplot2::geom_boxplot(width = 0.5, size = 0.8, outliers = FALSE, colour = "black") +
    ggplot2::theme_classic() +
    ggplot2::scale_colour_manual(values = rev(color_table$Colour)) +
    ggplot2::theme(legend.position = "none") +
    ggplot2::coord_flip() +
    ggplot2::scale_x_discrete(limits=rev) +
    ggplot2::scale_y_continuous(breaks = scales::breaks_pretty())
  return(p_out)
}

plot_style_col <- function(p_in){
  p_out <-
    p_in +
    ggplot2::geom_col(width = 0.8, colour = "black") +
    ggplot2::theme_classic() +
    ggplot2::scale_fill_manual(values = rev(color_table$Colour)) +
    ggplot2::theme(legend.position = "none") +
    ggplot2::coord_flip() +
    # ggplot2::scale_x_discrete(limits=rev) +
    ggplot2::scale_y_continuous(breaks = scales::breaks_pretty())
  return(p_out)
}

sig_for_leftjoin <-
  paste0('S',1:16) |>
  factor(levels = paste0('S',1:16)) |>
  tibble::as_tibble()

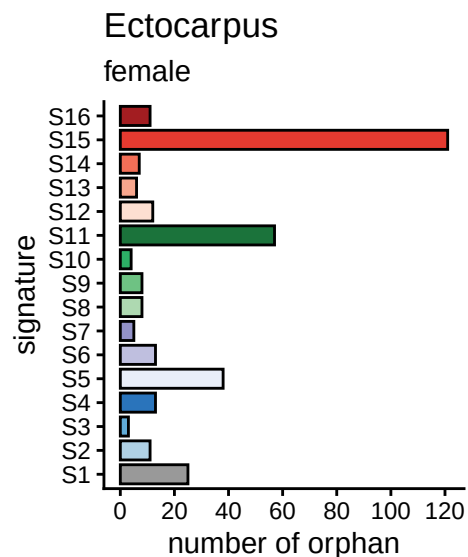
```

plotting signatures

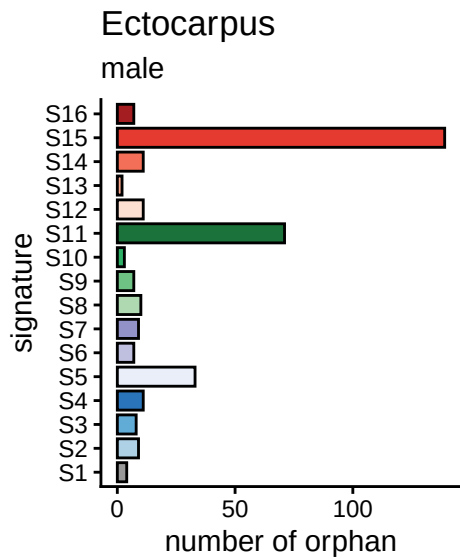
## hPTM signatures and number of genes

Ectocarpus

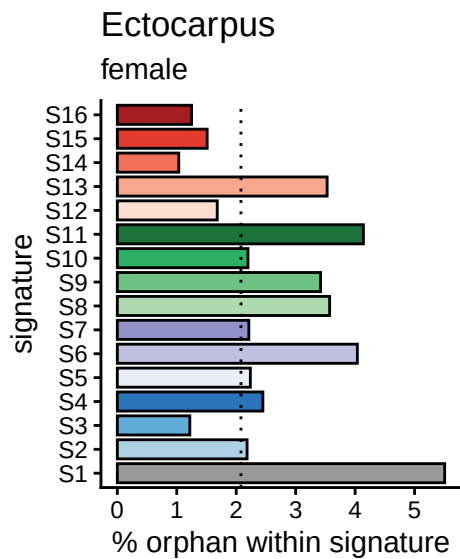
```
Ec_ps.filt_orphan.F <-  
  Ec_ps |>  
  preprocess_ps_sig_orphan(TPM_filt = 1) |>  
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Female, orphan)) |>  
  dplyr::distinct(Signature.Female, n_all.Female, n_orphan, orphan) |>  
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")  
  
Ec_ps.filt_orphan.M <-  
  Ec_ps |>  
  preprocess_ps_sig_orphan(TPM_filt = 1) |>  
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Male, orphan)) |>  
  dplyr::distinct(Signature.Male, n_all.Male, n_orphan, orphan) |>  
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")  
  
Ec_ps.filt_orphan.F.total <-  
  sum(Ec_ps.filt_orphan.F$orphan, na.rm = TRUE)/sum(Ec_ps.filt_orphan.F$n_all.Female)  
  
Ec_ps.filt_orphan.M.total <-  
  sum(Ec_ps.filt_orphan.M$orphan, na.rm = TRUE)/sum(Ec_ps.filt_orphan.M$n_all.Male)  
  
Ec_ps.filt_orphan.F |>  
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = orphan)) |>  
  plot_style_col() +  
  ggplot2::labs(x = "signature", y = "number of orphan",  
               title = "Ectocarpus", subtitle = "female")
```



```
Ec_ps.filt_orphan.M |>  
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = orphan)) |>  
  plot_style_col() +  
  ggplot2::labs(x = "signature", y = "number of orphan",  
               title = "Ectocarpus", subtitle = "male")
```

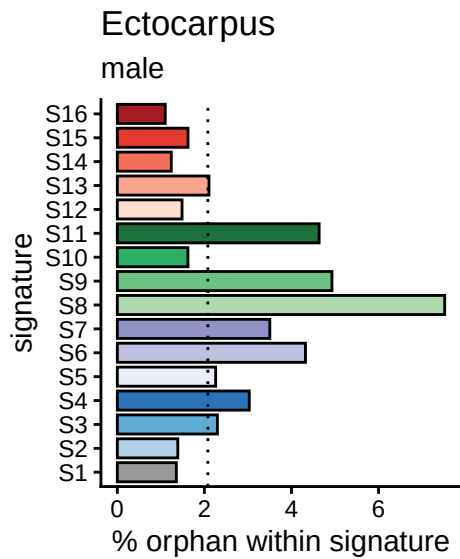


```
Ec_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = 100*orphan/n_all.Female)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Ec_ps.filt_orphan.F.total, colour = "black", linetype = 3)
ggplot2::labs(x = "signature", y = "% orphan within signature",
              title = "Ectocarpus", subtitle = "female")
```



```
# ggplot2::ggsave("figure/signature_orphan/Ec_orphan_rev2.pdf", width = 3, height = 3)

Ec_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = 100*orphan/n_all.Male)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Ec_ps.filt_orphan.M.total, colour = "black", linetype = 3)
ggplot2::labs(x = "signature", y = "% orphan within signature",
              title = "Ectocarpus", subtitle = "male")
```

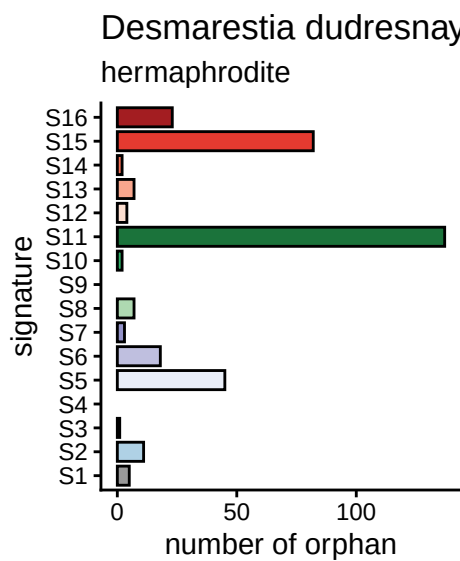


Desmarestia-dudresnayi

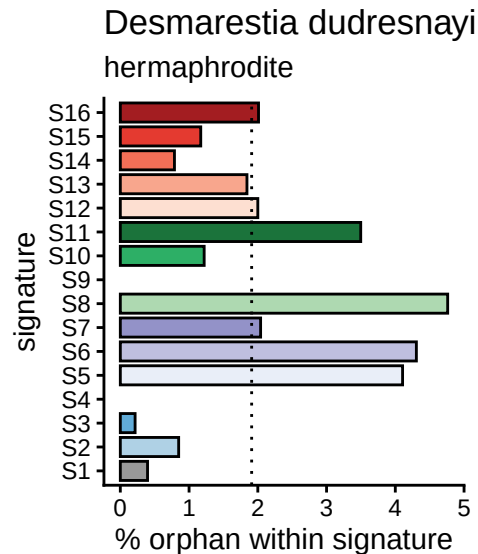
```
Ddud_ps.filt_orphan <-
  Ddud_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature, orphan)) |>
  dplyr::distinct(Signature, n_all, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

Ddud_ps.filt_orphan.total <-
  sum(Ddud_ps.filt_orphan$orphan, na.rm = TRUE)/sum(Ddud_ps.filt_orphan$n_all)

Ddud_ps.filt_orphan |>
  ggplot2::ggplot(aes(x = Signature, fill = Signature, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



```
Ddud_ps.filt_orphan |>
  ggplot2::ggplot(aes(x = Signature, fill = Signature, y = 100*orphan/n_all)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Ddud_ps.filt_orphan.total, colour = "black", linetype = 3)
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



```
# ggplot2::ggsave("figure/signature_orphan/Ddud_orphan_rev2.pdf", width = 3, height = 3)
```

Desmarestia-herbacea

```
Dher_ps.filt_orphan.F <-
  Dher_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Female, orphan)) |>
  dplyr::distinct(Signature.Female, n_all.Female, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

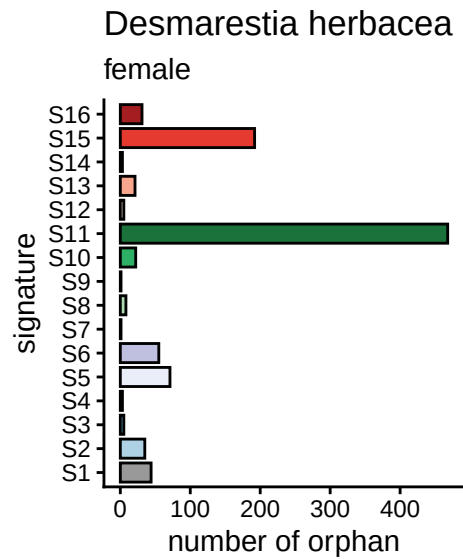
Dher_ps.filt_orphan.M <-
  Dher_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Male, orphan)) |>
  dplyr::distinct(Signature.Male, n_all.Male, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

Dher_ps.filt_orphan.F.total <-
  sum(Dher_ps.filt_orphan.F$orphan, na.rm = TRUE)/sum(Dher_ps.filt_orphan.F$n_all.Female)

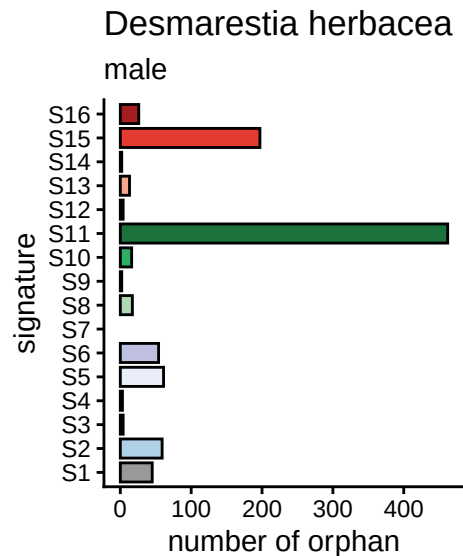
Dher_ps.filt_orphan.M.total <-
  sum(Dher_ps.filt_orphan.M$orphan, na.rm = TRUE)/sum(Dher_ps.filt_orphan.M$n_all.Male)

Dher_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
```

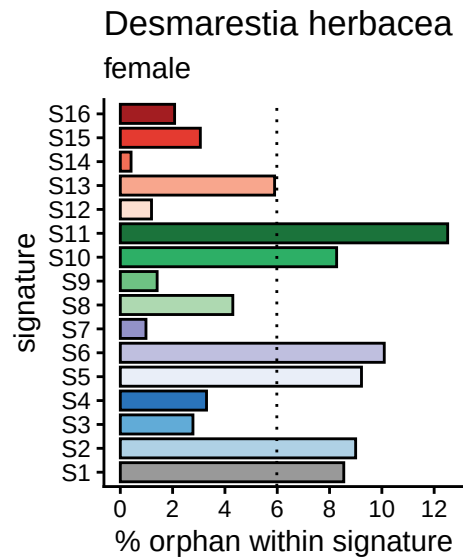
```
title = "Desmarestia herbacea", subtitle = "female")
```



```
Dher_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Desmarestia herbacea", subtitle = "male")
```

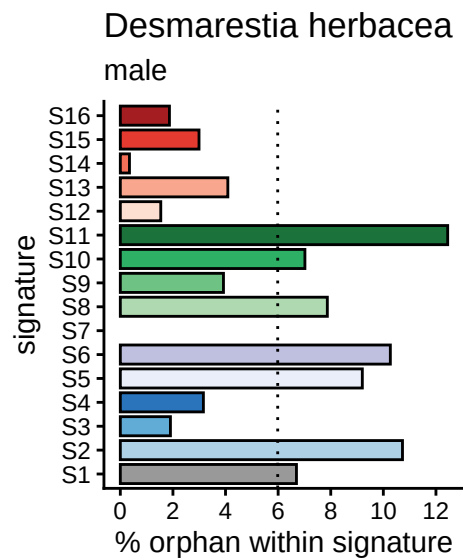


```
Dher_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = 100*orphan/n_all.Female)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Dher_ps.filt_orphan.F.total, colour = "black", linetype = "solid") +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Desmarestia herbacea", subtitle = "female")
```



```
# ggplot2::ggsave("figure/signature_orphan/Dher_orphan_rev2.pdf", width = 3, height = 3)
```

```
Dher_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = 100*orphan/n_all.Male)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Dher_ps.filt_orphan.M.total, colour = "black", linetype = "dashed") +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Desmarestia herbacea", subtitle = "male")
```



Undaria

```
Und_ps.filt_orphan.F <-
  Und_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Female, orphan)) |>
  dplyr::distinct(Signature.Female, n_all.Female, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

Und_ps.filt_orphan.M <-
```



```

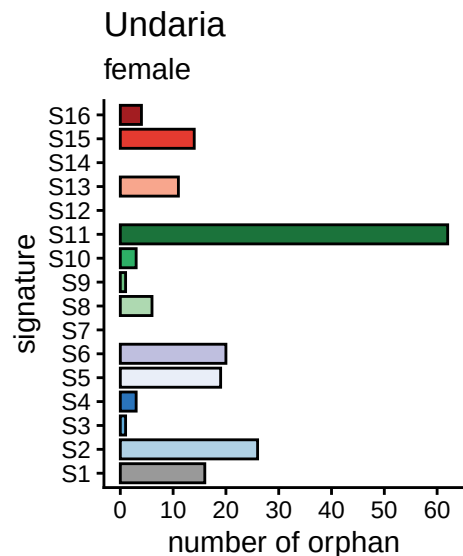
Und_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Male, orphan)) |>
  dplyr::distinct(Signature.Male, n_all.Male, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

Und_ps.filt_orphan.F.total <-
  sum(Und_ps.filt_orphan.F$orphan, na.rm = TRUE)/sum(Und_ps.filt_orphan.F$n_all.Female)

Und_ps.filt_orphan.M.total <-
  sum(Und_ps.filt_orphan.M$orphan, na.rm = TRUE)/sum(Und_ps.filt_orphan.M$n_all.Male)

Und_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Undaria", subtitle = "female")

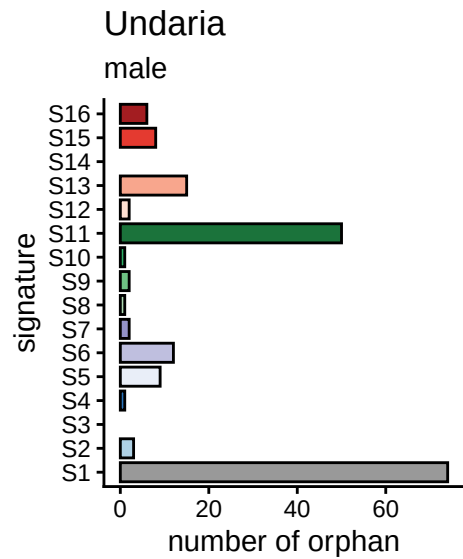
```



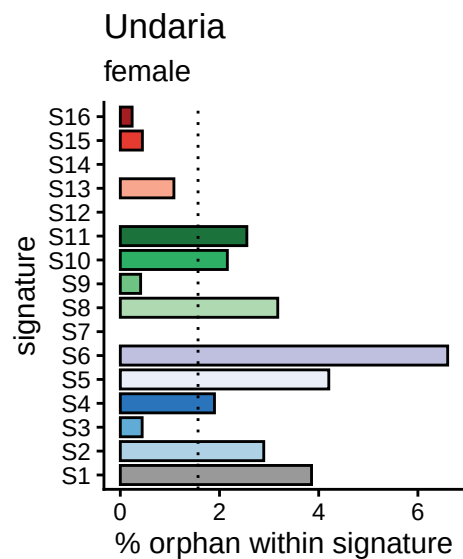
```

Und_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Undaria", subtitle = "male")

```

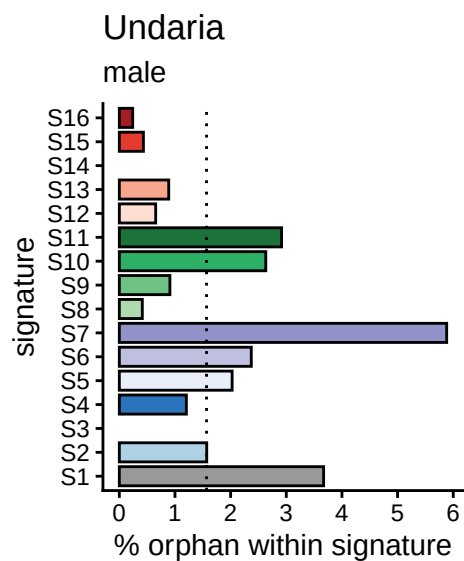


```
Und_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = 100*orphan/n_all.Female)) +
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Und_ps.filt_orphan.F.total, colour = "black", linetype = 3) +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Undaria", subtitle = "female")
```



```
# ggplot2::ggsave("figure/signature_orphan/Und_orphan_rev2.pdf", width = 3, height = 3)

Und_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = 100*orphan/n_all.Male)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Und_ps.filt_orphan.M.total, colour = "black", linetype = 3) +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Undaria", subtitle = "male")
```



Males removed since not all signatures are present (contains orphans).

Scytosiphon

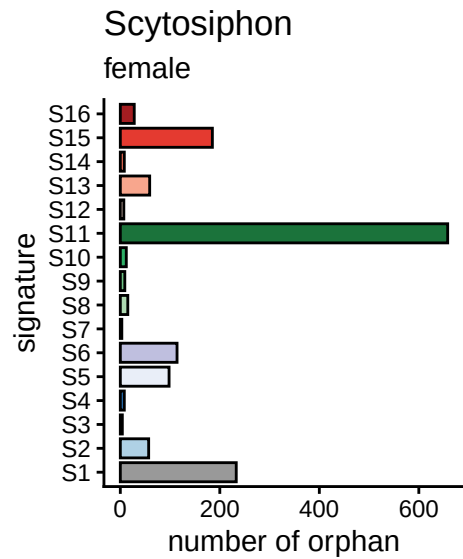
```
Spro_ps.filt_orphan.F <-
  Spro_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Female, orphan)) |>
  dplyr::distinct(Signature.Female, n_all.Female, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

Spro_ps.filt_orphan.M <-
  Spro_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature.Male, orphan)) |>
  dplyr::distinct(Signature.Male, n_all.Male, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

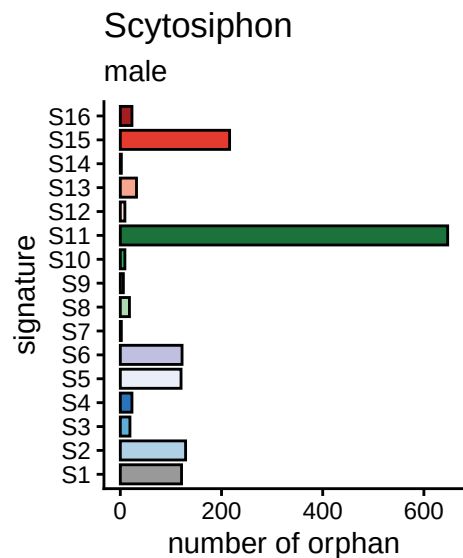
Spro_ps.filt_orphan.F.total <-
  sum(Spro_ps.filt_orphan.F$orphan)/sum(Spro_ps.filt_orphan.F$n_all.Female)

Spro_ps.filt_orphan.M.total <-
  sum(Spro_ps.filt_orphan.M$orphan)/sum(Spro_ps.filt_orphan.M$n_all.Male)

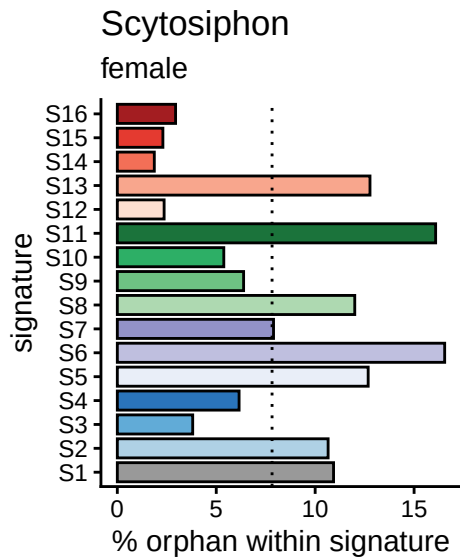
Spro_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Scytosiphon", subtitle = "female")
```



```
Spro_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
                title = "Scytosiphon", subtitle = "male")
```

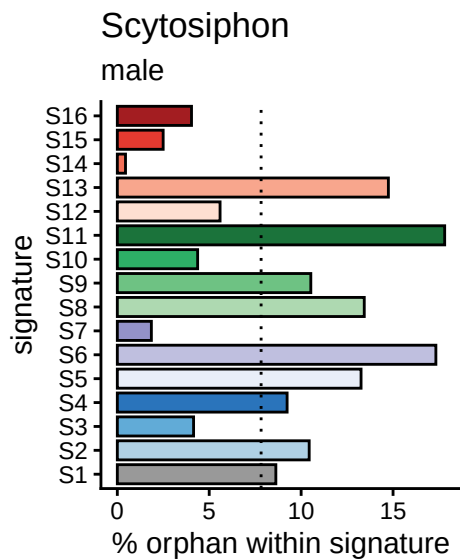


```
Spro_ps.filt_orphan.F |>
  ggplot2::ggplot(aes(x = Signature.Female, fill = Signature.Female, y = 100*orphan/n_all.Female)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Spro_ps.filt_orphan.F.total, colour = "black", linetype = "dashed") +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
                title = "Scytosiphon", subtitle = "female")
```



```
# ggplot2::ggsave("figure/signature_orphan/Spro_orphan_rev2.pdf", width = 3, height = 3)
```

```
Spro_ps.filt_orphan.M |>
  ggplot2::ggplot(aes(x = Signature.Male, fill = Signature.Male, y = 100*orphan/n_all.Male)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Spro_ps.filt_orphan.M.total, colour = "black", linetype = "dashed") +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Scytosiphon", subtitle = "male")
```



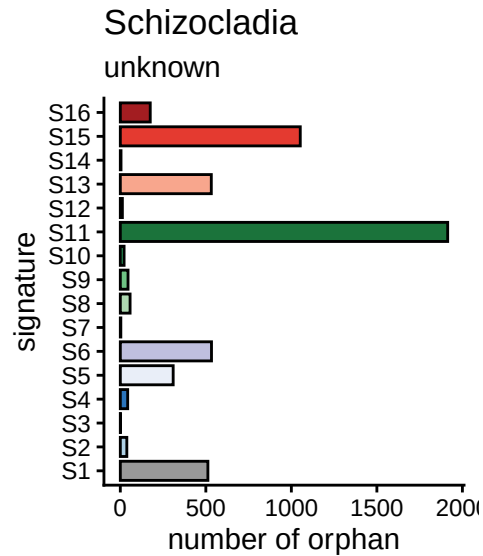
Schizocladia

```
Si_ps.filt_orphan <-
  Si_ps |>
  preprocess_ps_sig_orphan(TPM_filt = 1) |>
  dplyr::mutate(n_orphan = dplyr::n(), .by = c(Signature, orphan)) |>
  dplyr::distinct(Signature, n_all, n_orphan, orphan) |>
  tidyr::pivot_wider(names_from = "orphan", values_from = "n_orphan")

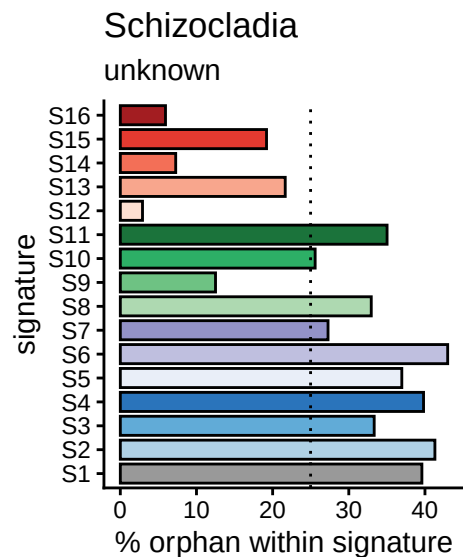
Si_ps.filt_orphan.total <-
```

```
sum(Si_ps.filt_orphan$orphan)/sum(Si_ps.filt_orphan$n_all)
```

```
Si_ps.filt_orphan |>
  ggplot2::ggplot(aes(x = Signature, fill = Signature, y = orphan)) |>
  plot_style_col() +
  ggplot2::labs(x = "signature", y = "number of orphan",
    title = "Schizocladia", subtitle = "unknown")
```



```
Si_ps.filt_orphan |>
  ggplot2::ggplot(aes(x = Signature, fill = Signature, y = 100*orphan/n_all)) |>
  plot_style_col() +
  ggplot2::geom_hline(yintercept = 100*Si_ps.filt_orphan$total, colour = "black", linetype = 3) +
  ggplot2::labs(x = "signature", y = "% orphan within signature",
    title = "Schizocladia", subtitle = "unknown")
```



```
# ggplot2::ggsave("figure/signature_orphan/Si_orphan_rev2.pdf", width = 3, height = 3)
```

Is it possible to combine plots?

```

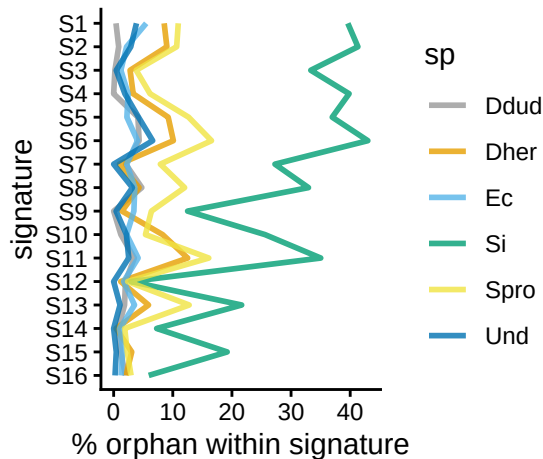
all_ps.filt_orphan.total <-
  tibble::tibble(
    sp = c("Ec", "Spro", "Und", "Ddud", "Dher", "Si"),
    p_total = c(
      Ec_ps.filt_orphan.F.total,
      Spro_ps.filt_orphan.F.total,
      Und_ps.filt_orphan.F.total,
      Ddud_ps.filt_orphan.total,
      Dher_ps.filt_orphan.F.total,
      Si_ps.filt_orphan.total
    )
  )
all_ps.filt_orphan <-
  Ec_ps.filt_orphan.F |>
  dplyr::mutate(sp = "Ec") |>
  dplyr::rename(Signature = Signature.Female,
    n_all = n_all.Female) |>
  rbind(
    Und_ps.filt_orphan.F |> dplyr::mutate(sp = "Und") |>
      dplyr::rename(Signature = Signature.Female,
        n_all = n_all.Female)
  ) |>
  rbind(
    Ddud_ps.filt_orphan |> dplyr::mutate(sp = "Ddud")
  ) |>
  rbind(
    Dher_ps.filt_orphan.F |> dplyr::mutate(sp = "Dher") |>
      dplyr::rename(Signature = Signature.Female,
        n_all = n_all.Female)
  ) |>
  rbind(
    Spro_ps.filt_orphan.F |> dplyr::mutate(sp = "Spro") |>
      dplyr::rename(Signature = Signature.Female,
        n_all = n_all.Female)
  ) |>
  rbind(
    Si_ps.filt_orphan |> dplyr::mutate(sp = "Si")
  ) |>
  dplyr::left_join(all_ps.filt_orphan.total) |>
  dplyr::mutate(orphan = dplyr::case_when(
    is.na(orphan) ~ 0,
    .default = orphan
  )) |>
  dplyr::mutate(p_orphan = orphan/n_all,
    p_diff = p_orphan - p_total)

cbPalette <- c("#999999", "#E69F00", "#56B4E9", "#009E73", "#F0E442", "#0072B2", "#D55E00", "#CC79A7")

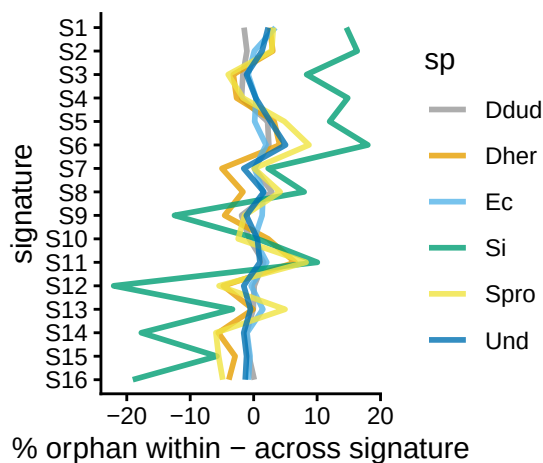
all_ps.filt_orphan |>
  ggplot2::ggplot(aes(x = Signature, colour = sp, group = sp, y = 100*p_orphan)) +
  ggplot2::geom_line(size = 1, alpha = 0.8) +
  ggplot2::labs(x = "signature", y = "% orphan within signature") +
  ggplot2::theme_classic() +
  ggplot2::coord_flip() +

```

```
ggplot2::scale_x_discrete(limits=rev) +
ggplot2::scale_y_continuous(breaks = scales::breaks_pretty()) +
ggplot2::scale_colour_manual(values=cbPalette)
```



```
all_ps.filter_orphan |>
  ggplot2::ggplot(aes(x = Signature, colour = sp, group = sp, y = 100*p_diff)) +
  ggplot2::geom_line(size = 1, alpha = 0.8) +
  ggplot2::labs(x = "signature", y = "% orphan within - across signature") +
  ggplot2::theme_classic() +
  ggplot2::coord_flip() +
  ggplot2::scale_x_discrete(limits=rev) +
  ggplot2::scale_y_continuous(breaks = scales::breaks_pretty()) +
  ggplot2::scale_colour_manual(values=cbPalette)
```



## session info

```
devtools::session_info()
```

```
## - Session info -----
## setting value
## version R version 4.4.0 (2024-04-24)
## os      macOS 15.5
## system  x86_64, darwin20
```



```

## ui      X11
## language (EN)
## collate en_US.UTF-8
## ctype   en_US.UTF-8
## tz      Europe/Berlin
## date    2025-10-29
## pandoc  3.4 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/ (via rmark
##
## - Packages -----
## package      * version date (UTC) lib source
## beeswarm      0.4.0   2021-06-01 [1] CRAN (R 4.4.0)
## cachem        1.1.0   2024-05-16 [1] CRAN (R 4.4.0)
## cellranger    1.1.0   2016-07-27 [1] CRAN (R 4.4.0)
## cli           3.6.5   2025-04-23 [1] CRAN (R 4.4.1)
## devtools      2.4.5   2022-10-11 [1] CRAN (R 4.4.0)
## dichromat     2.0-0.1 2022-05-02 [1] CRAN (R 4.4.0)
## digest        0.6.37  2024-08-19 [1] CRAN (R 4.4.1)
## dplyr          * 1.1.4   2023-11-17 [1] CRAN (R 4.4.0)
## ellipsis      0.3.2   2021-04-29 [1] CRAN (R 4.4.0)
## evaluate      1.0.5   2025-08-27 [1] CRAN (R 4.4.1)
## farver        2.1.2   2024-05-13 [1] CRAN (R 4.4.0)
## fastmap       1.2.0   2024-05-15 [1] CRAN (R 4.4.0)
## forcats       * 1.0.0   2023-01-29 [1] CRAN (R 4.4.0)
## fs            1.6.6   2025-04-12 [1] CRAN (R 4.4.1)
## generics      0.1.4   2025-05-09 [1] CRAN (R 4.4.1)
## ggbeeswarm    0.7.2   2023-04-29 [1] CRAN (R 4.4.0)
## ggplot2       * 4.0.0   2025-09-11 [1] CRAN (R 4.4.1)
## ggrastr       * 1.0.2   2023-06-01 [1] CRAN (R 4.4.0)
## glue          1.8.0   2024-09-30 [1] CRAN (R 4.4.1)
## gtable        0.3.6   2024-10-25 [1] CRAN (R 4.4.1)
## hms           1.1.3   2023-03-21 [1] CRAN (R 4.4.0)
## htmltools     0.5.8.1 2024-04-04 [1] CRAN (R 4.4.0)
## htmlwidgets  1.6.4   2023-12-06 [1] CRAN (R 4.4.0)
## httpuv        1.6.16  2025-04-16 [1] CRAN (R 4.4.1)
## knitr         1.50    2025-03-16 [1] CRAN (R 4.4.1)
## later         1.4.4   2025-08-27 [1] CRAN (R 4.4.1)
## lifecycle     1.0.4   2023-11-07 [1] CRAN (R 4.4.0)
## lubridate     * 1.9.4   2024-12-08 [1] CRAN (R 4.4.1)
## magrittr      2.0.4   2025-09-12 [1] CRAN (R 4.4.1)
## memoise       2.0.1   2021-11-26 [1] CRAN (R 4.4.0)
## mime          0.13    2025-03-17 [1] CRAN (R 4.4.1)
## miniUI        0.1.2   2025-04-17 [1] CRAN (R 4.4.1)
## pillar        1.11.1  2025-09-17 [1] CRAN (R 4.4.1)
## pkgbuild      1.4.4   2024-03-17 [1] CRAN (R 4.4.0)
## pkgconfig     2.0.3   2019-09-22 [1] CRAN (R 4.4.0)
## pkgload       1.4.0   2024-06-28 [1] CRAN (R 4.4.0)
## profvis       0.4.0   2024-09-20 [1] CRAN (R 4.4.1)
## promises      1.3.3   2025-05-29 [1] CRAN (R 4.4.1)
## purrr         * 1.1.0   2025-07-10 [1] CRAN (R 4.4.1)
## R6            2.6.1   2025-02-15 [1] CRAN (R 4.4.1)
## RColorBrewer  1.1-3   2022-04-03 [1] CRAN (R 4.4.0)
## Rcpp          1.1.0   2025-07-02 [1] CRAN (R 4.4.1)
## readr         * 2.1.5   2024-01-10 [1] CRAN (R 4.4.0)
## readxl        * 1.4.3   2023-07-06 [1] CRAN (R 4.4.0)

```

```

## remotes      2.5.0    2024-03-17 [1] CRAN (R 4.4.0)
## rlang        1.1.6    2025-04-11 [1] CRAN (R 4.4.1)
## rmarkdown    2.29     2024-11-04 [1] CRAN (R 4.4.1)
## rstudioapi    0.17.1   2024-10-22 [1] CRAN (R 4.4.1)
## S7           0.2.0    2024-11-07 [1] CRAN (R 4.4.1)
## scales       1.4.0    2025-04-24 [1] CRAN (R 4.4.1)
## sessioninfo  1.2.2    2021-12-06 [1] CRAN (R 4.4.0)
## shiny        1.9.1    2024-08-01 [1] CRAN (R 4.4.0)
## stringi      1.8.7    2025-03-27 [1] CRAN (R 4.4.1)
## stringr      * 1.5.2    2025-09-08 [1] CRAN (R 4.4.1)
## tibble       * 3.3.0    2025-06-08 [1] CRAN (R 4.4.1)
## tidyr        * 1.3.1    2024-01-24 [1] CRAN (R 4.4.0)
## tidyselect   1.2.1    2024-03-11 [1] CRAN (R 4.4.0)
## tidyverse    * 2.0.0    2023-02-22 [1] CRAN (R 4.4.0)
## timechange    0.3.0    2024-01-18 [1] CRAN (R 4.4.0)
## tinytex      0.53     2024-09-15 [1] CRAN (R 4.4.1)
## tzdb         0.5.0    2025-03-15 [1] CRAN (R 4.4.1)
## urlchecker   1.0.1    2021-11-30 [1] CRAN (R 4.4.0)
## usethis      3.1.0    2024-11-26 [1] CRAN (R 4.4.1)
## vctrs        0.6.5    2023-12-01 [1] CRAN (R 4.4.0)
## vipor        0.4.7    2023-12-18 [1] CRAN (R 4.4.0)
## withr        3.0.2    2024-10-28 [1] CRAN (R 4.4.1)
## xfun         0.52     2025-04-02 [1] CRAN (R 4.4.1)
## xtable       1.8-4    2019-04-21 [1] CRAN (R 4.4.0)
## yaml        2.3.10   2024-07-26 [1] CRAN (R 4.4.0)
##
## [1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
##
## -----

```